

Predicting Weekly Dengue Cases from Climate History

Why changing the representation of time beat adding model depth
DengAI competition – final project report

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October 2026

Abstract

Dengue is a mosquito-borne disease whose weekly case counts depend on weather that happened weeks or months earlier. In the DengAI competition we forecast weekly cases for San Juan (Puerto Rico) and Iquitos (Peru) from climate and vegetation measurements. This report is the record of an iterative investigation rather than a description of one final model. XGBoost and CatBoost first looked stronger than a forest on training and chronological validation, yet the hidden leaderboard reversed that ranking. That failure led us to a lower-variance, city-specific 500-tree system with at least five samples per leaf (22.5 hidden MAE), and then to the more important discovery that all tree models were missing outbreak peaks. We traced the problem to delayed climate effects, selected feature- and city-specific history windows, compressed their 461/380 raw MLP inputs into 180 stable multiscale summaries, and finally appended four Partial Least Squares (PLS) scores for San Juan. Hidden MAE improved from 26.7 to 22.5, 18.8, 16.6, and finally **15.9**. The main result is not merely the score: careful error analysis and a better representation of time were more valuable than additional model complexity.

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1 Problem and motivation

Dengue fever is transmitted by *Aedes* mosquitoes. Temperature, humidity and rainfall influence mosquito breeding, survival and the incubation of the virus, but each with a delay: a wet period now tends to translate into more infections several weeks later. Public-health agencies must decide *in advance* where to place vector-control teams, hospital beds and diagnostic supplies, so a forecast of next weeks' case counts that is available before the cases appear has direct operational value: it can shift effort from reaction to prevention.

The DengAI competition [1] frames this as a supervised regression problem. For each of two cities we are given weekly records of satellite vegetation indices (NDVI), station temperature and rainfall, and reanalysis climate variables, together with the number of reported cases (`total_cases`) for the training period. The task is to predict `total_cases` for a later, hidden period and the score is the mean absolute error

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{g}_i|. \quad (1)$$

What makes the task hard.

- The two cities have very different climates and case scales, so one model may not fit both.
- Case counts are heavy-tailed: most weeks are quiet, a few weeks are large outbreaks, and absolute error is dominated by how well those peaks are anticipated.
- Weather affects cases with city- and variable-specific delays, so the way the past is presented to a model matters.
- Training data are small (a few hundred to about a thousand weeks per city) and time-ordered, so naive validation is optimistic.
- True case counts for the test period are unavailable, so *previous cases cannot be used as predictors*; the model must forecast from climate history alone.

Executive Summary. Our work was a sequence of failures and questions, each created by the previous result (Figure 1). Boosting won locally but lost on the hidden period, so we reduced tree variance. The resulting forest still missed outbreak peaks, so we stopped tuning the learner and inspected the time series. That investigation suggested delayed infection-cycle effects, which led to feature-specific climate histories. Those histories helped but were too large for only 1456 rows, which led to multiscale compression and finally a small supervised PLS augmentation. The final two representation changes produced the largest gains.

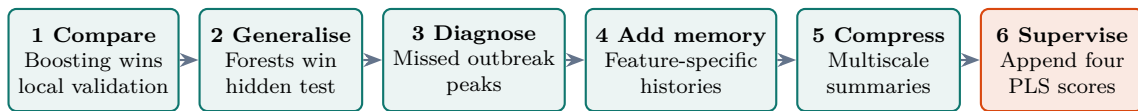


Figure 1: The experiment map. A disappointing result or visible failure mode triggered each next experiment.

2 Data

2.1 Loading and splitting

We load the three CSV files with `pandas`: features, labels, and the test features. Training features and labels are joined on the key (`city`, `year`, `weekofyear`) with a one-to-one check, giving a table

with 1456 rows. The inputs X are the weekly climate columns and the target y is `total_cases`. The test keys are verified to match the submission template exactly, so predictions can be written in the required order.

Table 1: Dataset overview.

	San Juan (SJ)	Iquitos (IQ)
Training weeks	936	520
Test weeks (hidden)	260	156
Training period	1990–2008	2000–2010
Median weekly cases	19	5
Mean weekly cases	34.2	7.6
Maximum weekly cases	461	116

The 24 test columns comprise three key columns, the date, four NDVI indices, and 16 weather series from stations and reanalysis. Tree models use NDVI summaries as well as weather. The final MLP expands only the 16 consistently defined weather series, plus calendar phase: NDVI has substantially more missingness and did not provide stable lag-window evidence. This is why “24 columns” becomes 16 temporally engineered signals rather than 24 lagged variables.

2.2 Two cities, two target distributions

Figure 2 shows the weekly case series and their distributions. San Juan has rare, very large outbreaks (maximum 461, median 19), while Iquitos has smaller counts with many near-zero weeks (maximum 116, median 5). Both distributions are strongly right-skewed.

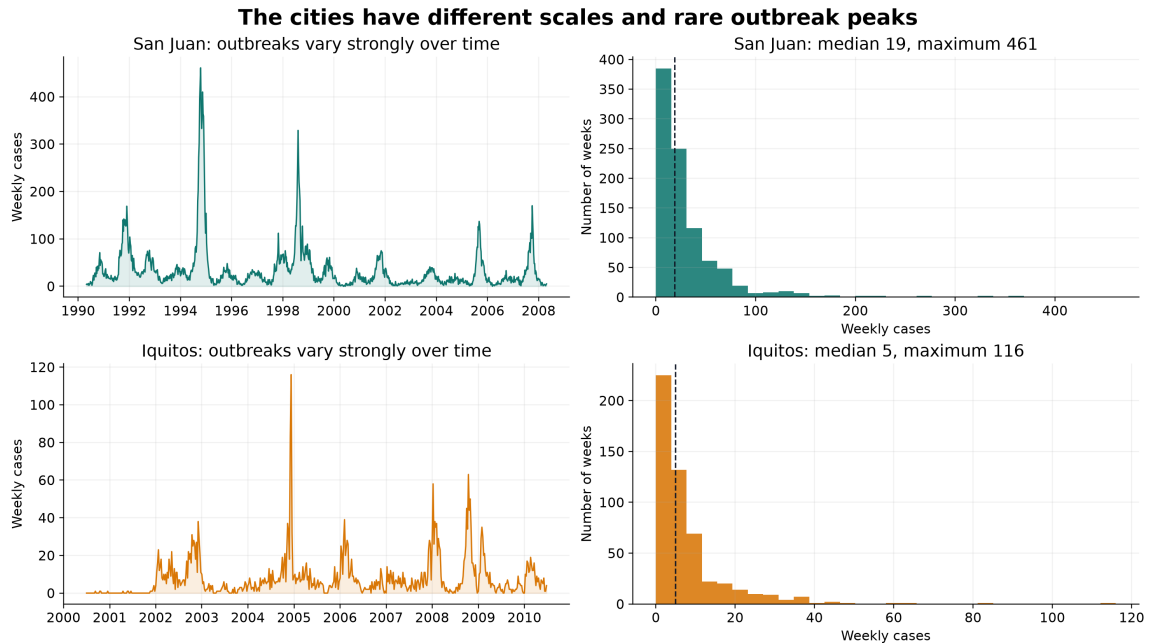


Figure 2: Weekly cases per city (left) and their distributions with the median marked (right). The scales differ by roughly a factor of four and both are dominated by rare peaks.

Decision. The cities differ in scale, climate and outbreak behaviour, so we train and tune a *separate model for each city*. Because the metric is absolute error and outbreaks are rare, we also

report error separately on outbreak weeks (weeks above a city-specific high-case threshold) in addition to the mean.

3 Exploratory analysis

3.1 Seasonality helps but does not decide

Figure 3 plots the mean, median and interquartile range of cases per calendar week. San Juan peaks in the second half of the year (highest mean around week 40) and Iquitos shows a late-year peak (around week 50) with an additional early-year bump. However, the interquartile bands are wide: the same week of the year can have very different outcomes in different years.

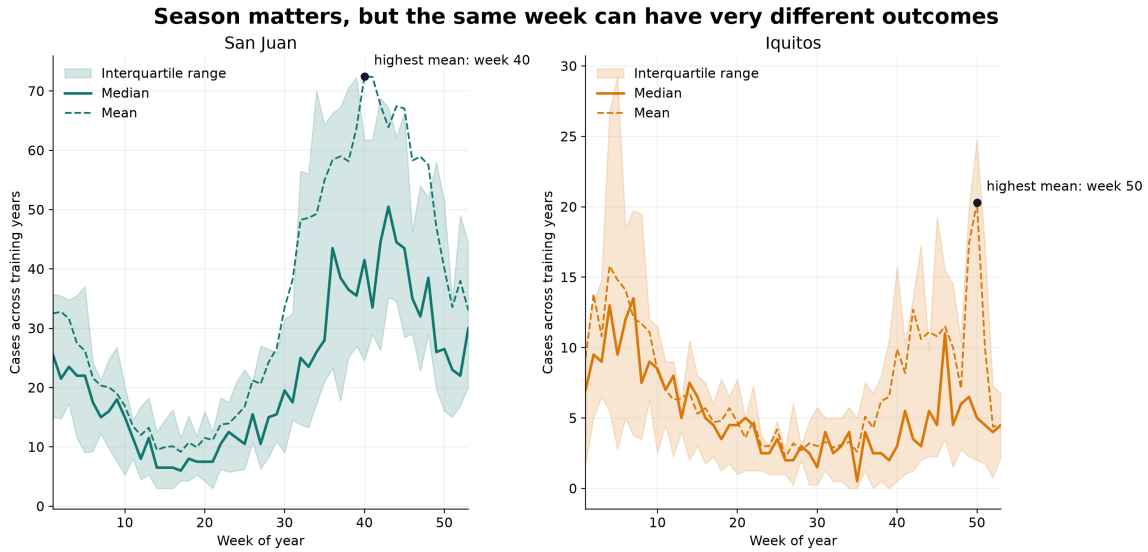


Figure 3: Cases by calendar week across training years. The seasonal signal is real, but the wide band shows that calendar week alone does not determine the count.

Decision. The model needs the *calendar phase* (as continuous sine/cosine terms so that week 52 and week 1 are neighbours) *and* recent climate conditions.

3.2 Missing values

Missing values are rare but not negligible: the training features contain 548 missing cells. Figure 4 lists the variables missing in at least 1% of training weeks. The northeast NDVI index in San Juan is the worst case (about 20%), followed by several Iquitos station temperature variables (about 7%).

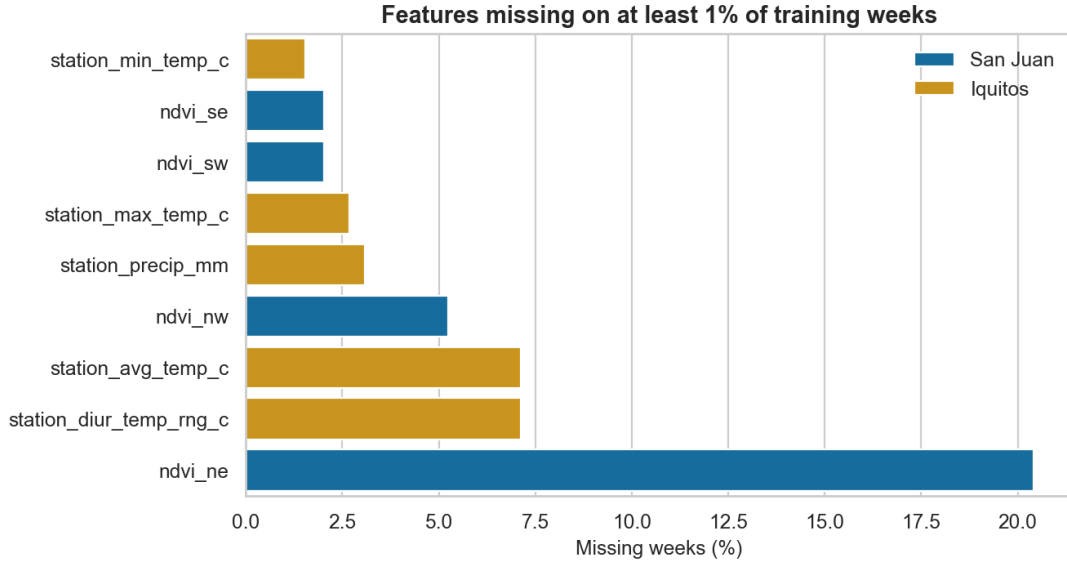


Figure 4: Share of training weeks with a missing value, by variable and city.

3.3 Case counts have memory; climate memory is variable-specific

The autocorrelation of weekly cases (left of Figure 5) decays over roughly 15–20 weeks, much more slowly in San Juan than in Iquitos. To determine the relevant climate history, we screened candidate history lengths from 3 to 51 weeks. To ensure strict forecasting validity, we evaluated these lengths using the chronological expanding-window folds described in Section 5.1. A decision-tree screen scored the consecutive window $[x_{t-L+1}, \dots, x_t]$ for each length L . As shown in Figure 10, the resulting heatmap selected very different window lengths: for example 34 weeks of relative humidity in San Juan but only 7 in Iquitos.

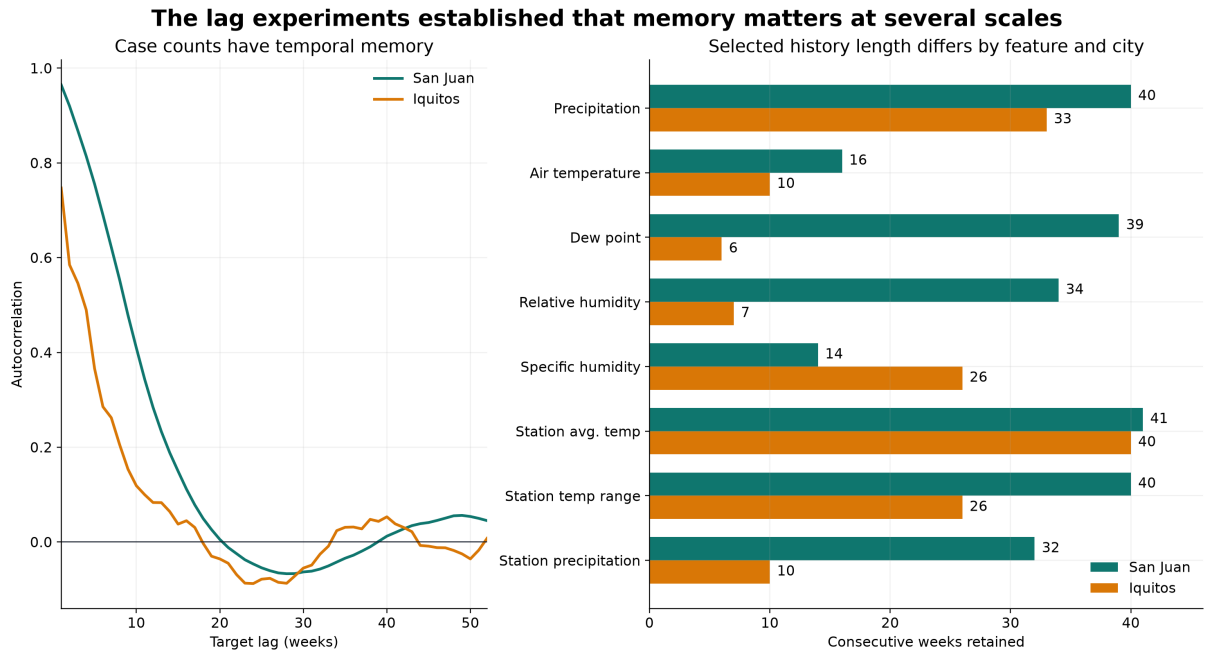


Figure 5: Left: autocorrelation of cases. Right: history length (number of consecutive weeks retained) chosen by the screening step, per weather variable and city.

Caveat. The original ten-fold procedure used for this screen was not fully forecasting-safe, so

exact values such as 39 versus 40 weeks should not be read as biological constants. We use the screen only as evidence that memory is long, variable-specific and city-specific.

4 Preprocessing and feature engineering

Every transformation below is motivated by one of the findings above. Test-time features use only climate observations available at the model’s recorded history endpoint. For exact experimental compatibility, the reproduction notebook also retains the original two-row alignment and circular context used to construct the earliest training windows; we document this implementation quirk rather than silently changing the submitted model.

Table 2: Preprocessing steps and the reason for each.

Step	What we do	Why
Missing values, forests	Median imputer fitted on the training fold	The median is robust and fold-fitting avoids leakage
Missing values, boosting	XGBoost/CatBoost native missing-value routing	Each split learns a default direction for unavailable values; no global fill value is required
Missing values, MLP	Linear interpolation in time order	Weather is smooth over a week; interpolation avoids breaking history windows
Scaling (tree models)	None: raw units are retained	Split order is unchanged by monotone rescaling; scaling adds no benefit to XGBoost, CatBoost, Random Forest or Extra Trees
Scaling (MLP)	z -scores of weather columns using San Juan statistics; week of year min–max scaled	Variables are in different units; networks are scale-sensitive
Season	$\sin / \cos$ of $2\pi \text{ week}/52$ (and 4π for tree models, four harmonics)	Smooth, continuous yearly cycle across the new-year boundary
Target	Predict cases directly with an absolute-error loss; Iquitos trees use $\log(1+y)$	MAE is the competition metric; log stabilises Iquitos’ small counts for forests
Per-city models	One model and one set of hyper-parameters per city	Different scales, histories and regimes (Section 3)

4.1 Exactly what XGBoost and CatBoost received

The boosting comparison used the same information as the confirmed forest, so it was a model comparison rather than a different-feature comparison. We sorted each city chronologically, removed identifiers that should not drive prediction (`city`, `date` and `target`), and removed calendar year for the boosting candidates because extrapolating a numerical year trend into a three- or five-year test block was unsafe. All remaining inputs were numeric; CatBoost was not given categorical columns.

For each weekly row, the design matrix contained current climate and NDVI measurements, eight Fourier coordinates (four annual harmonics), NDVI mean and spatial spread, two temperature ranges, a temperature–humidity interaction, nine weather variables at lags 1, 2, 4, 8, 12, 16, and past-only means over 2, 4, 8, 12, 16 weeks. Rolling means were shifted by one week. Consequently, “past mean 4” at week t used $t - 1$ through $t - 4$, never the current value and never a future value. Previous dengue counts were excluded because true case labels are unavailable throughout

the hidden horizon.

Table 3: Preprocessing and regularisation in the first model comparison.

Model	Input handling	Main controls used in the experiment
XGBoost	Same engineered numeric matrix; missing values handled by the learner; no standardisation	500 boosting rounds, learning rate 0.03, depth 2 or 3, minimum child weight 10, row subsampling 0.85, feature subsampling 0.8, L1 0.1, L2 10; MAE and Poisson objectives tested
CatBoost	Same engineered numeric matrix; native missing-value routing; no categorical features and no standardisation	500 iterations, learning rate 0.03, depth 4 or 6, L2 leaf penalty 10, Bernoulli subsampling 0.85, random strength 0.5; MAE and Poisson losses tested
Random Forest / Extra Trees	Fold-fitted median imputation; no standardisation	500 trees and at least five samples per leaf; city-specific target/criterion described in Section 6.2

The boosting targets were the observed case counts: Poisson candidates enforced a count-compatible objective and MAE candidates matched the leaderboard loss. We did not normalize the target. Only the Iquitos forest applied $\log(1 + y)$ and transformed its predictions back with $\exp(\hat{z}) - 1$. Every learned imputation rule and every fitted model used only the earlier part of each chronological fold.

4.2 Three feature sets

(A) Tree features. For each week: current climate; four annual Fourier harmonics (eight seasonal coordinates); NDVI mean and spread over the four locations; temperature ranges; a temperature $\times$ humidity interaction; climate at lags 1, 2, 4, 8, 12, 16; and trailing means over 2, 4, 8, 12, 16 weeks (computed with a one-week shift so the current row cannot leak into its own summary).

(B) Raw lag windows. For each weather variable v and city c we keep a window of $L_{v,c}$ consecutive past values (not a single lag). Concatenating all windows gives one flat vector: 461 inputs for San Juan and 380 for Iquitos. A 40-week window keeps all 40 consecutive values, not only the value at lag 40.

(C) Multiscale summaries. For each of the 16 weather variables we compute 11 statistics from the previous 52 weeks, and add four seasonal terms:

1. the current (latest) value;
2. means over 2, 4, 8, 13, 26, 52 weeks (climate level at several timescales);
3. standard deviations over 4 and 13 weeks (recent and medium-term variability);
4. the 4-week mean minus the 13-week mean (short-term conditions relative to baseline);
5. a 13-week trend, $(x_t - x_{t-12})/12$ (direction and rate of change);
6. sin, cos of 2π week/52 and of 4π week/52 (cyclic seasonality).

The size is $16 \times 11 + 4 = \mathbf{180}$ inputs. The final San Juan model appends four PLS scores, giving 184 inputs; the final Iquitos model retains its 380 raw-lag inputs.

5 Validation protocol and metric

5.1 Why chronological validation

A random split places neighbouring outbreak weeks on both sides of the split. The model can then “interpolate” an outbreak from adjacent weeks that would never be available in a real forecast, giving an optimistic score. The hidden test period lies strictly after the training period, so validation must have the same direction.

We use **expanding-window folds**: fold k trains on all weeks before a cut point and is evaluated on the next 52 weeks (Figure 6). All preprocessing that learns from data (imputation, scaling, any projection) is fitted inside the training window of each fold.

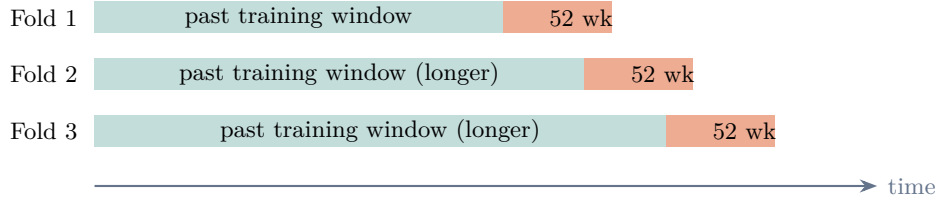


Figure 6: Expanding folds: the model never sees data later than the period it is scored on.

5.2 Metric

We use MAE because it is the competition metric and is less dominated by single extreme weeks than squared error. We report three numbers per configuration: the *mean* MAE across folds, the *worst-fold* MAE (stability), and the *outbreak-week* MAE (the weeks that matter most for public health). The same folds are used for every model so that differences measure the model and not the split.

5.3 Baselines

Figure 7 compares a historical-mean predictor, a week-of-year climatology (the median or mean for that calendar week) and a tuned tree ensemble on the same 312 weeks per city. Climatology is very hard to beat in Iquitos (6.8 vs. 7.1 for the trees), whereas in San Juan climate features cut the error from 25.1 to 15.7. Figure 8 shows why San Juan is difficult: climatology predicts a smooth seasonal hump every year, while actual outbreaks are irregular.

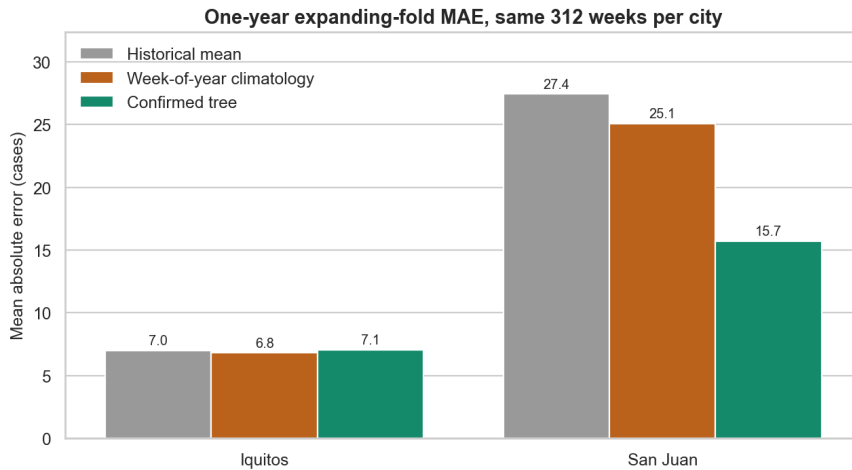


Figure 7: One-year expanding-fold MAE on identical 312 validation weeks per city.

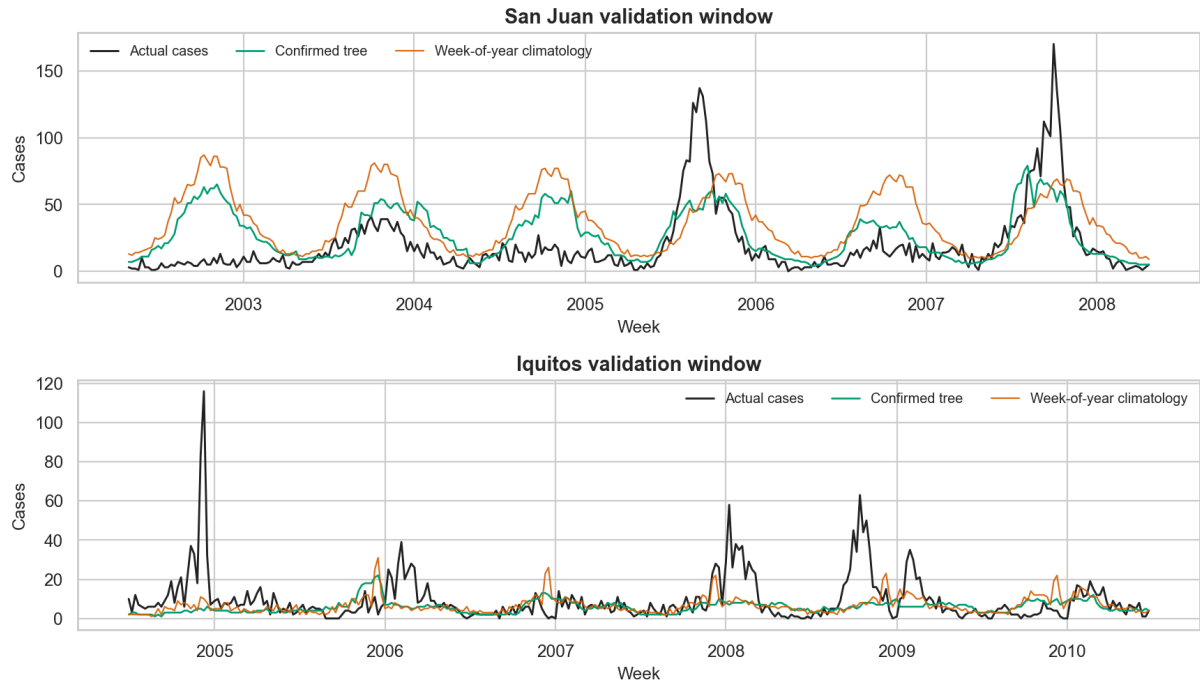


Figure 8: Validation windows: actual cases, a confirmed tree model, and week-of-year climatology. Peaks of the largest outbreaks are underestimated.

6 Models and experiments

6.1 Stage 0: the first ranking reversal

Our first serious comparison used XGBoost [2], CatBoost [3] and a Random-Forest family baseline [4]. The early training and validation runs favoured the boosted learners: sequentially correcting residuals let them fit small irregularities that a bagged forest averaged away. The archived controlled experiment confirms the local part of that observation. It selected a 20% XGBoost contribution for Iquitos and a 40% CatBoost contribution for San Juan, and both the four-fold selection score and the six-fold report score improved over the forest system (Table 4).

Table 4: The local ranking that did not survive the hidden period. The boosted row is a city-specific blend, not a claim that one boosted model won both cities.

Candidate	4-fold local MAE	6-fold local MAE	Hidden-test outcome
City-specific forest	10.091	11.393	22.5; retained
Selected boosted blend	9.788	11.061	Worse than forest; rejected

The leaderboard reversed the local ranking: the simpler forest submission generalized better. We cannot prove whether this was ordinary model-selection overfitting, temporal distribution shift, or both, because the hidden labels are unavailable. The important engineering conclusion was operational: a small local MAE advantage on a few dependent time folds was not enough evidence to trust a more adaptive model. This changed our selection rule from “take the lowest validation mean” to “also inspect fold variance, peak amplitude and behaviour on a test-length horizon.”

6.2 Stage 1: lower-variance forest ensembles (hidden MAE 22.5)

We next focused on variance. Random Forest fits trees to bootstrap samples and random feature subsets. Extra Trees goes further: for candidate features it randomizes the split thresholds before choosing a split. This extra randomisation makes the individual trees less similar. If T trees have variance σ^2 and average pairwise correlation ρ , the variance of their average is approximately

$$\text{Var}(\bar{f}) = \frac{\sigma^2}{T} [1 + (T - 1)\rho]. \quad (2)$$

With $T = 500$, reducing correlation ρ is at least as important as adding more trees. Extra Trees can therefore reduce ensemble variance, although excessive randomisation can increase bias. We controlled the other major source of variance by requiring at least five training examples in every leaf.

The selected city-specific system used the causal tree features of set (A):

- San Juan: 500-tree Extra Trees [5] with Poisson splitting;
- Iquitos: 500-tree Random Forest trained on $\log(1 + y)$;
- minimum five samples per leaf to avoid very specific rules.

How the ensemble prediction is formed. This is not a vague vote among model names. Within one city, every tree produces a real-valued case forecast and the forest averages the 500 outputs:

$$\hat{y}_{c,t} = \frac{1}{500} \sum_{b=1}^{500} f_{c,b}(x_{c,t}). \quad (3)$$

For Iquitos the average is formed on the $\log(1 + y)$ scale and then inverted; for San Juan the trees predict the count directly with the Poisson split criterion. Predictions are clipped at zero and rounded to an integer. We do *not* average San Juan and Iquitos models together: the San Juan model predicts San Juan rows, the Iquitos model predicts Iquitos rows, and the two blocks are concatenated in the official submission order.

We also tested explicit forest–boosting blends,

$$\hat{y} = (1 - w)\hat{y}_{\text{forest}} + w\hat{y}_{\text{boost}}, \quad (4)$$

with $w \in \{0, 0.1, \dots, 1.0\}$. Local validation selected $w = 0.2$ with XGBoost for Iquitos and $w = 0.4$ with CatBoost for San Juan. Those blends are the second row of Table 4; because their hidden result was worse, they were experiments, not the submitted 22.5 tree system.

Trees capture non-linear thresholds (“warm *and* humid after several wet weeks”), lags express delayed effects, rolling means express accumulated exposure, and Fourier terms express season. This reduced hidden MAE from 26.7 to 22.5. But the validation forecast in Figure 8 exposed the next problem: the forest followed the seasonal level while repeatedly cutting off the largest peaks.

Residual Analysis & Feature Engineering

We inspected the largest absolute residuals week by week. Several outbreak weeks had a sudden increase in cases even though the weather in that same row was not unusually extreme. The model had learned an overly instantaneous mapping—“this week’s weather predicts this week’s cases”—while the real process includes mosquito development, viral incubation and reporting delays. A rainfall or humidity episode can therefore matter weeks before the observed case peak.

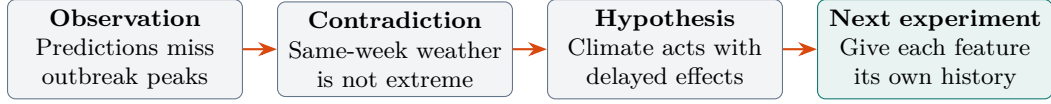


Figure 9: Error analysis converted a visible failure mode into the lag-window experiment.

This was the project’s main pivot. We stopped asking only “which learner wins?” and asked “what information is missing from the input?” The answer was not one universal lag for all variables, but a city- and feature-specific history.

6.3 Stage 2: an MLP with lagged inputs (hidden MAE $19.3 \rightarrow 18.8$)

To test that hypothesis, we screened candidate history lengths from 3 through 51 weeks separately for every weather variable and city. For a candidate length L , a small decision tree received that variable’s entire consecutive window $[x_{t-L+1}, \dots, x_t]$ and was scored by MAE. **As shown in Figure 10**, the resulting heatmap did not select the features themselves; it compared how much history of each feature was useful.

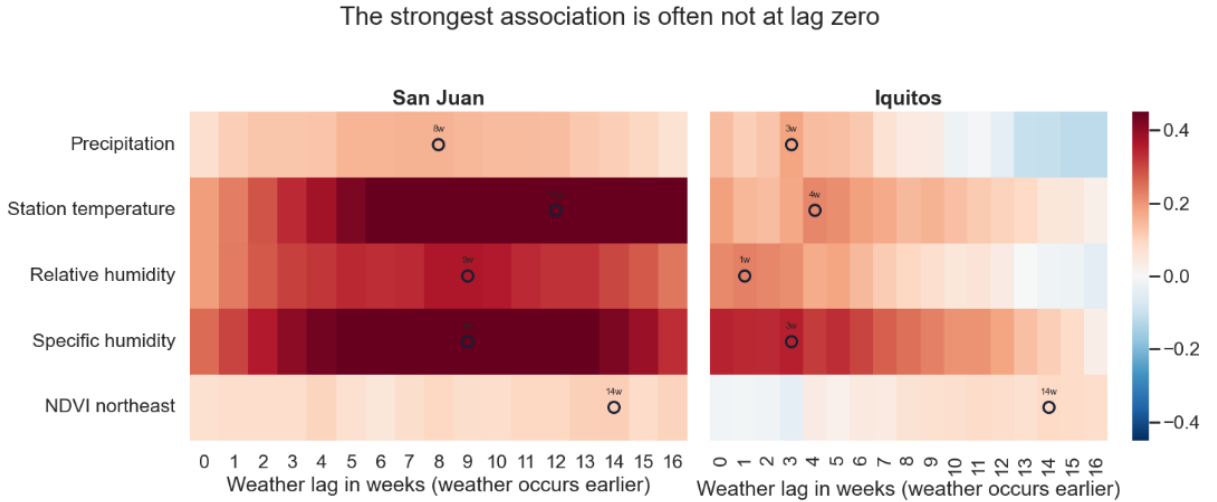


Figure 10: Heatmap of MAE scores for candidate history lengths across weather variables. The darkest regions indicate the lowest error, which guided our selection of specific consecutive window lengths (e.g., 34 weeks for relative humidity in San Juan) rather than using a universal lag.

A 40-week choice therefore means all 40 consecutive values, not only lag 40.

The strongest region of the heatmap was chosen for each feature and city, producing very different windows—for example, relative humidity used 34 weeks in San Juan and 7 in Iquitos. Because the original screen used ordinary ten-fold validation and nearby windows were often nearly tied, we treat exact values such as 39 versus 40 as engineering choices, not biological constants. The reliable insight was that climate memory is long, variable-specific and city-specific.

We then gave a neural network this structured memory. Concatenating the selected windows produced 461 inputs for San Juan and 380 for Iquitos. The network is a three-layer perceptron with SELU activations [6], dropout, a linear output and an absolute-error loss, trained with RMSprop and a plateau learning-rate decay. Figure 11 shows the two city architectures.

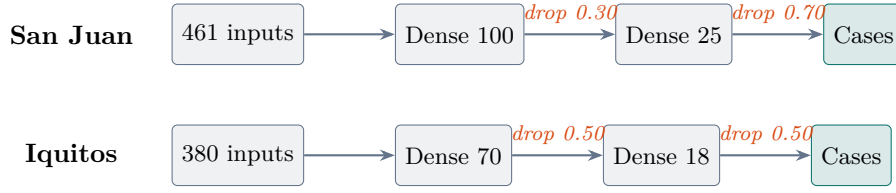


Figure 11: Raw-history MLPs after temporal tuning. In the multiscale model the input is 180 for both cities and the San Juan network keeps the same $100 \rightarrow 25$ hidden layers.

Table 5: Consecutive history retained by the raw-lag MLP. The entries are window lengths in weeks, not isolated lag positions.

Feature	SJ	IQ	Feature	SJ	IQ
Precipitation amount	40	33	Satellite precipitation	40	33
Reanalysis air temperature	16	10	Specific humidity	14	26
Reanalysis average temperature	15	4	Reanalysis diurnal range	21	34
Reanalysis dew point	39	6	Station average temperature	41	40
Reanalysis maximum temperature	12	41	Station diurnal range	40	26
Reanalysis minimum temperature	21	40	Station maximum temperature	37	39
Reanalysis precipitation	30	3	Station minimum temperature	26	25
Relative humidity	34	7	Station precipitation	32	10
Week of year	3	3			

Tuning. Under temporal cross-validation, the useful change was a *narrower second hidden layer*, not a deeper network:

Table 6: Effect of tuning the second hidden layer (mean temporal-CV MAE).

	Second layer	Before	After
San Juan	50 $\rightarrow$ 25	22.02	20.64
Iquitos	35 $\rightarrow$ 18	8.17	7.91

This gave a hidden MAE of 18.8 for seed 42; averaged over three seeds it is 19.5, a reminder that a single seed overstates performance.

Remaining problem. The MLP improved the hidden result, supporting the time-lag hypothesis, but it created a statistical mismatch. The full dataset has only 1456 labels, and separate-city training leaves 936 San Juan labels and 520 Iquitos labels. Against those sample sizes, the network receives 461 and 380 highly correlated coordinates per week. In San Juan the first layer alone has 46,100 input weights. The model must discover simple concepts such as “recent mean”, “variability” and “trend” from individual positions, and shifting a wet episode by one week moves its information into many different coordinates. The representation was informative but inefficient.

Why eleven summaries per weather variable. We did not choose eleven as a magic number. We listed the distinct questions the raw trajectory should answer while remaining small: What is the latest level? What has the average level been over short, monthly, two-month, quarterly, half-year and annual horizons? How volatile has it been recently and over a quarter? Is the recent month above the quarterly baseline? Is the variable rising or falling? These questions become one current value, six means (2, 4, 8, 13, 26, 52 weeks), two standard deviations (4, 13

weeks), one short-minus-medium contrast, and one 13-week slope: $1 + 6 + 2 + 1 + 1 = 11$. The horizons align with interpretable calendar scales and overlap deliberately, allowing the MLP to compare short- and long-term exposure without seeing all 52 raw positions.

6.4 Stage 3: multiscale summaries (hidden MAE 16.6)

We therefore replaced raw coordinates with the 180 summaries of feature set (C) and kept *the same MLP, training procedure and seed*. Under identical chronological folds (three folds, three seeds), summaries lower the mean and worst-fold errors in both cities (Table 7). San Juan improves on mean, worst and outbreak error; Iquitos improves on mean and worst error, while its outbreak MAE rises slightly (from 23.2 to 23.8).

Table 7: Same tuned MLP, chronological validation, three folds and three seeds: raw lag windows vs. multiscale summaries (lower is better).

City	Representation	Mean MAE	Worst MAE	Outbreak MAE
SJ	Raw lags	20.635	31.942	57.095
SJ	Multiscale summaries	17.726	27.346	55.844
IQ	Raw lags	7.908	12.481	23.244
IQ	Multiscale summaries	7.588	10.596	23.795

Weighted across the two cities by the competition weighting, the mean MAE improves from 15.862 to 13.924. On the hidden leaderboard, the error fell from 18.8 to 16.6. Replacing only San Juan and replacing both cities both scored 16.6, so the gain comes from San Juan and the Iquitos representation change is approximately neutral at leaderboard precision.

Is it compression or just more information?

If the summaries helped by adding information, then raw lags *plus* summaries should be best. In a two-fold ablation it was the worst (Figure 12): San Juan MAE was 17.86 for raw histories, **12.61** for summaries and 18.43 for raw plus summaries. The improvement came from *discarding redundant coordinates and exposing stable temporal structure*. Summaries also cut the San Juan parameter count from 48,751 to 20,651 without flattening outbreaks (maximum test prediction rose from 174 to 200).

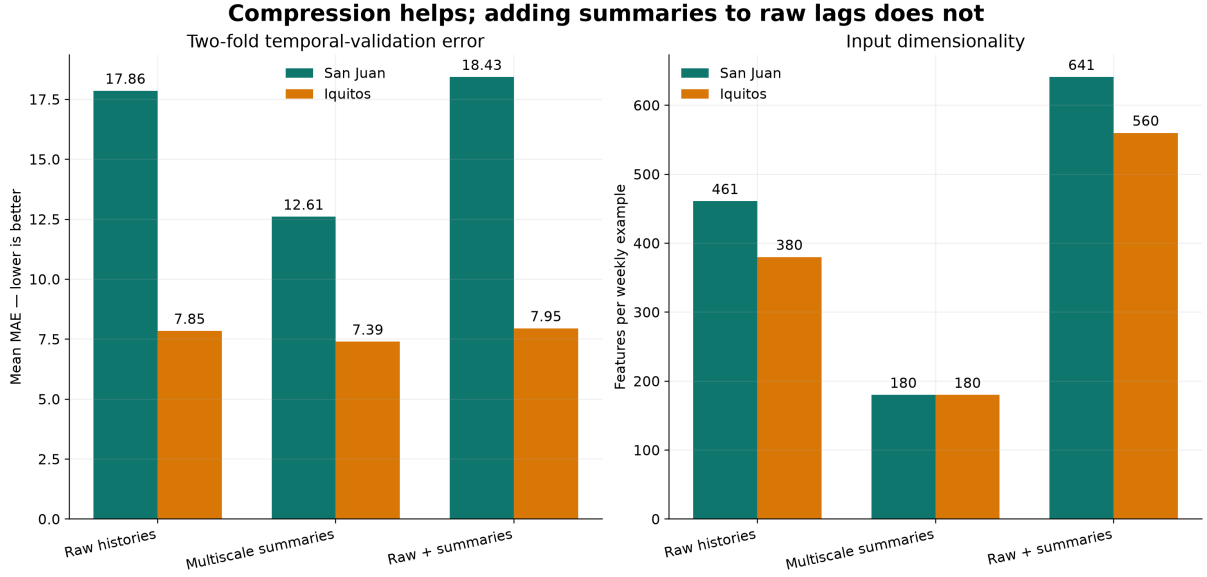


Figure 12: Left: two-fold temporal-validation MAE for three input representations. Right: number of inputs per weekly example. Compression helps; adding summaries to raw lags does not.

6.5 Stage 4: four PLS scores (final hidden MAE 15.9)

Even the 180 summaries remain correlated: for example, 8-, 13- and 26-week temperature means often move together. Partial Least Squares (PLS) [7] creates a small number of directions chosen to covary with the target. Let $X \in \mathbb{R}^{n \times 180}$ be the multiscale matrix and $y \in \mathbb{R}^n$ the weekly cases. A simplified PLS component $t_k = Xw_k$ uses a weight vector that solves

$$w_k = \arg \max_{\|w\|=1} \text{Cov}(Xw, y)^2, \quad (5)$$

then deflates X and y before extracting the next direction. Unlike PCA, PLS uses y : it asks which directions in climate history are associated with cases, not merely which directions explain climate variance.

Append, do not replace. PLS-only inputs discarded useful nonlinear and seasonal detail, while many components were unstable. The selected design keeps all 180 summaries and appends only four standardized scores:

$$180 \text{ summaries} + 4 \text{ PLS scores} \longrightarrow 100 \longrightarrow 25 \longrightarrow 1.$$

The four scores add only 2.2% to the input dimension. They act as supervised hints about useful cross-variable combinations; the network can still use every original rainfall, temperature, humidity and season statistic directly. PLS is applied to San Juan, where the multiscale representation produced the clear leaderboard gain; Iquitos remains on the 380-input raw-lag model.

Leakage-safe protocol. Because PLS uses the target, it is refitted inside every chronological fold:

1. fit interpolation/scaling parameters on the earlier training window;
2. build 180 summaries and fit PLS using only that window and its targets;
3. standardize PLS scores with training-window statistics;

4. freeze the projection, transform the next 52 weeks, and train/evaluate the 184-input MLP.

Fitting PLS once before splitting would let future labels shape the representation and invalidate validation.

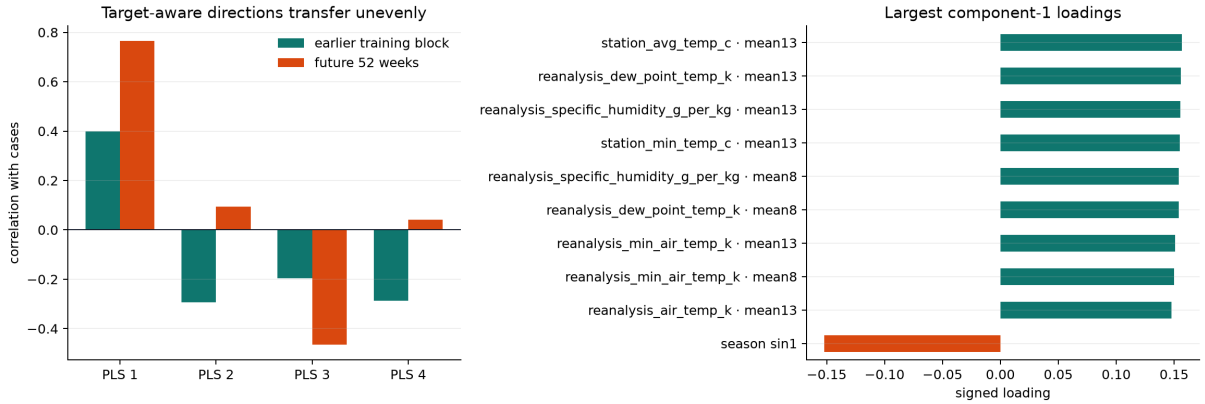


Figure 13: Leakage-safe PLS diagnostics on DengAI San Juan data. Left: correlations between the four training-fitted scores and cases in the earlier training block and the future 52-week block. Right: largest signed loadings in component 1. The components transfer unevenly, which is why only four are appended and the original summaries are retained.

The local comparison was close relative to fold variance, so PLS was a bounded representation experiment rather than a claim of a decisive cross-validation win. The hidden score improved from 16.6 to the final **15.9** without increasing network depth or changing Iquitos. This is evidence that a small target-aware augmentation can complement stable historical summaries.

7 Results and analysis

7.1 The leaderboard journey

Table 8 and Figure 14 summarise the hidden-test MAE of every stage.

Table 8: Hidden-test MAE across stages (lower is better).

Stage	Representation	Predictor	MAE
First baseline	Earlier feature pipeline	Baseline model	26.7
Tree system	Current climate, sparse lags, rolling means, season	SJ Extra Trees; IQ Random Forest	22.5
Raw-history MLP	Feature-specific contiguous windows	3-layer MLP	19.3
Improved dropout	Same windows	3-layer MLP	19.1
Tuned raw-history MLP	461 SJ / 380 IQ raw values	Tuned MLP, seed 42	18.8
Multiscale MLP	180 summaries	Same tuned MLP	16.6
Narrow multiscale	Same 180 summaries	48 → 12 MLP	18.1 (rejected)
PLS-augmented multiscale	180 summaries + 4 PLS scores (SJ)	Same tuned MLP	15.9

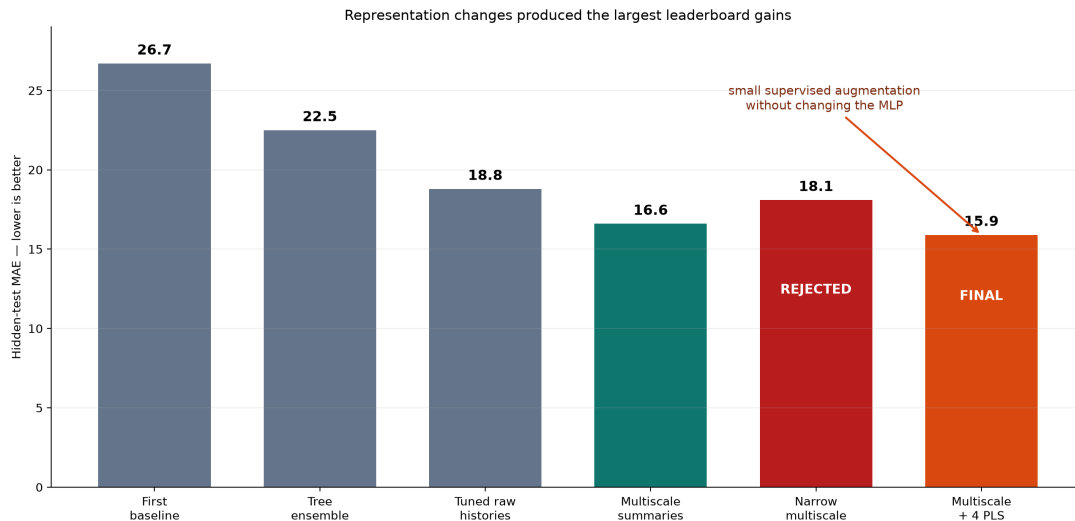


Figure 14: Hidden-test MAE by stage. The 16.6 model is the pre-PLS parent; four target-aware scores produce the final 15.9 result.

7.2 The rejected experiments were part of the result

The score sequence can look deceptively direct in hindsight. In practice, much of the work was spent establishing what *not* to trust. We preserved the most informative negative results in Table 9; they explain why the final system is a hybrid rather than the winner of one large hyperparameter search.

Table 9: Experiments that did not enter the final submission and the information they contributed.

Experiment	What happened	Lesson carried forward
XGBoost/CatBoost and forest blends	Better training/development behaviour and lower local MAE, but worse hidden generalisation	Temporal validation is evidence, not proof; preserve a simple fallback and check prediction amplitude
Forest hyperparameter and component search	Several mixtures won on development origins but not on the untouched recent origin	Prefer small, stable gains; use test-length chronological horizons
Negative-binomial, volatility, recency and quantile corrections	Plausible count/time-series ideas did not improve robustly enough to replace the confirmed forest	A sophisticated output model cannot recover climate history that is absent from the input
Three-seed raw-lag MLP ensemble	Averaging seeds scored 19.5, while seed 42 scored 18.8	Averaging highly correlated models can shrink rare peaks instead of adding useful diversity
Raw lags plus summaries	More inputs were worse than summaries alone (18.43 vs. 12.61 in the two-fold ablation)	The gain was compression and inductive bias, not simply more information
Narrow 48 → 12 multiscale MLP	Won local architecture search by 0.224 MAE but lost hidden test by 1.5 MAE	Freeze the proven network when a searched local gain is smaller than fold variance

This effort was worthwhile because each negative result narrowed the diagnosis. Model-family tuning exposed generalisation risk; peak-by-peak residual inspection exposed missing temporal context; the raw-plus-summary ablation proved that compression, not feature count, caused

the large improvement. The final score is therefore the end of an evidence chain, not a lucky architecture choice.

7.3 Local validation is evidence, not proof

After shrinking the input from 461 to 180, a local architecture search preferred a narrower $48 \rightarrow 12$ network by only 0.224 validation MAE, and its outbreak error was already worse. On the hidden test it was 1.5 MAE *worse* (18.1 vs. 16.6; Figure 15). A small win after searching many candidates is likely selection noise. The representation survived hidden testing; the architecture difference did not. We therefore froze the $100 \rightarrow 25$ San Juan network and concentrated on controlled ablations of the representation.

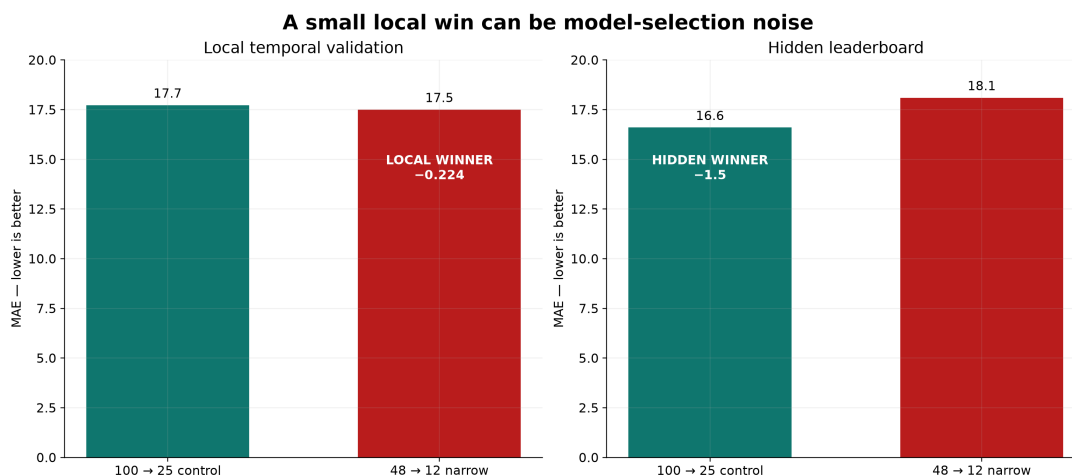


Figure 15: The $48 \rightarrow 12$ network looked better locally (-0.224) but was worse on the hidden test ($+1.5$).

7.4 Final submission and reproducibility boundary

The final submission is a city hybrid. San Juan uses 180 summaries plus four fold-fitted, standardized PLS scores, followed by hidden layers $100 \rightarrow 25$, dropout 0.3/0.7, learning rate 0.01, seed 42, and MAE loss. Iquitos keeps the tuned 380-input raw-history MLP ($70 \rightarrow 18$, dropout 0.5/0.5). The competition requires non-negative integer predictions; the notebook clips at zero, applies the recorded city-specific integer rule, restores the official template order, asserts all 416 keys, and prints a SHA-256 digest.

The notebook distinguishes three kinds of evidence. The 15.9 value is the observed hidden-leaderboard score; hidden labels are unavailable locally. Chronological hold-out MAE is a development measurement and is never presented as hidden-test performance. Exact byte identity is checked against the uploaded CSV when that reference artifact is present. This separation prevents a saved notebook output or a locally good fold from being misreported as a leaderboard result.

Before PLS, we also tested integer conversion for the 16.6 parent (Figure 16). Nearest-integer rounding raised 128 of 260 San Juan predictions by one case and improved that model’s local hold-out MAE from 16.635 to 16.519. This was a small post-processing check, not the source of the final PLS gain.

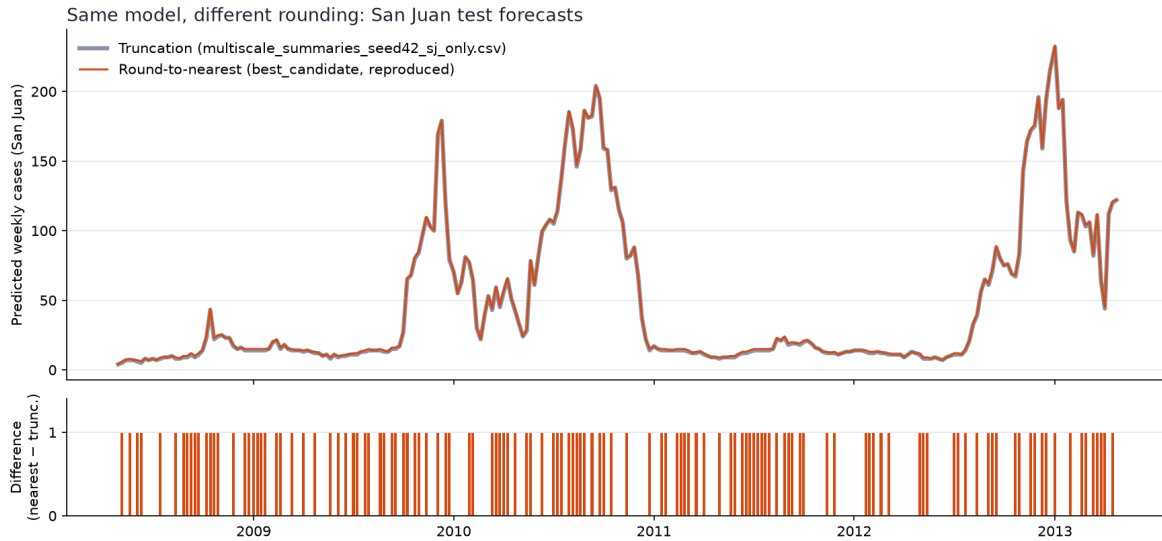


Figure. The two submissions share the same multiscale-feature MLP (seed 42) and identical Iquitos predictions; they differ only in how San Juan predictions are rounded to integers. Round-to-nearest raises 128 of 260 San Juan weeks by exactly +1 case (never lowers any), removing the downward bias of truncation (-0.49 cases/week; totals 12,582 vs 12,454).

Figure 16: San Juan test forecasts of the final model with truncation vs. round-to-nearest (top) and their difference (bottom).

7.5 Error analysis

- **Outbreaks changed the representation.** In San Juan the outbreak-week MAE (≈ 56) is more than three times the mean MAE (≈ 18) (Table 7). The early trees predicted the seasonal envelope but missed abrupt peaks whose same-week weather looked ordinary. That residual pattern, not a generic desire for more features, motivated delayed climate windows and then multiscale summaries. The final model times major rises more convincingly, although peak amplitude remains uncertain.
- **Fold variance is large.** The worst San Juan fold (27.3) is much worse than the mean (17.7): performance depends on which year is held out.
- **Single seeds mislead.** The 18.8 result had a three-seed mean of 19.5.
- **Iquitos is near its floor.** Climatology is as good as complex models on Iquitos validation weeks (≈ 7), consistent with its small, noisy counts.

8 Discussion: does the model solve the problem?

What the model achieves. Relative to the first baseline, hidden MAE falls by about 40% ($26.7 \rightarrow 15.9$) and relative to the tree system by about 29%. Because mean weekly cases are about 34 in San Juan and 7.6 in Iquitos, the model forecasts seasonal level and major rises usefully, but it is not accurate enough to provide exact counts for individual outbreak weeks.

Real-world applicability. The inputs are routinely collected climate series and the forecast uses *no* previous case counts, so it can run without a delay in case reporting and complements a surveillance system that does. Plausible uses are early warning for vector-control timing, staffing and supply planning at the level of weeks to months. We did not measure the economic or health impact of such decisions; that would require deployment data and is outside the evidence we have.

Limitations.

- Two cities and about 20 years of data: generalisation to other locations is untested.
- Climate-only inputs miss immunity, serotype mix, human mobility and control campaigns that drive outbreak size.
- Model selection on a few chronological folds is noisy (Section 7.2); the hidden leaderboard is a single test period.
- Point forecasts only; decision makers would also need prediction intervals.
- The reproduced pipeline preserves a historical two-row alignment/circular training-context quirk for compatibility; a clean reimplementaion should compare this explicitly with a strictly chronological alignment.

Lessons.

1. A model that wins training and local validation can still lose on a later hidden regime. The XGBoost/CatBoost reversal is why we kept chronological horizons, amplitude checks and simple fallbacks.
2. Averaging 500 decorrelated trees and requiring five samples per leaf gave a strong small-data baseline, but lower estimator variance could not repair missing temporal information.
3. The same small network improved from 18.8 to 16.6 when the *description of the past* improved: level, duration, variability, direction and season.
4. Four supervised PLS directions improved the hidden result again to 15.9 without replacing the robust summaries or increasing network depth.
5. More features are not better features: raw plus summaries was the worst of three options. The most valuable work was error analysis followed by controlled ablation.

9 Future work

- Repeat the four-component PLS result over more temporal cut points and seeds, and archive the exact submitted CSV and hash alongside the score.
- Quantify uncertainty (quantile loss or ensembles over seeds) to give outbreak risk bands rather than a single count.
- Evaluate more folds and repeated seeds before accepting architecture changes smaller than the fold-to-fold spread.
- Test the representation on other cities or other vector-borne diseases.

10 Conclusion

We forecast weekly dengue cases in two cities from climate history. The path was not “try a larger model until the score improves.” XGBoost and CatBoost first won locally and lost on the hidden period. A lower-variance, city-specific 500-tree system with minimum leaf size five recovered to 22.5, but its missed outbreak peaks revealed that the input was too instantaneous. Feature-specific contiguous histories let an MLP exploit delayed climate effects (18.8), then exposed a second problem: 461/380 correlated inputs for only 936/520 city labels. Replacing raw coordinates with 180 multiscale summaries improved hidden MAE to 16.6; appending four

leakage-safe, target-aware PLS scores for San Juan produced the final **15.9**. Neither gain required a deeper network. The durable lesson is that *error analysis identified the missing idea, and a stable representation of time mattered more than model complexity*.

Author Contributions

- **Duy Chien Nguyen:** Conducted data preprocessing, developed the baseline models, fine-tuned model hyperparameters, implemented Partial Least Squares (PLS) to improve statistical representations, made the presentation slides, and wrote the report.
- **Dang Quy Nguyen:** Conducted data preprocessing, devveloped and improved the tree-based methodologies, led feature engineering efforts, built the lagged feature representations and the Multi-Layer Perceptron (MLP) architecture, made the presentation slides, and wrote the report.

A Reproducibility

- Data: DengAI training/test CSV files from the competition site.
-
- **Code Repository:** The complete source code, trained models, and reproducible notebooks are available on GitHub at <https://github.com/dangnq2501/DengAI>.
- **Presentation Video:** A recorded video of our project presentation can be viewed at https://youtu.be/Et3oHyoI_VI.
- The hidden MAE 15.9 is a leaderboard observation. Exact byte-level reproduction is asserted only when the uploaded reference CSV is stored at the path documented by the notebook.
- Figures in this report were produced by the project’s plotting scripts and notebooks and are included in the `figures/` folder.

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